

CS311 MCQs for Final Term

XML DOM MCQs

What does DOM stand for in XML DOM?

- A) Document Object Model
- B) Data Object Model
- C) Document Oriented Model
- D) Data Oriented Model

Answer: A) Document Object Model

Which of the following is NOT a method of the XML DOM?

- A) getElementById()
- B) createElement()
- C) appendChild()
- D) getNodeValue()

Answer: D) getNodeValue()

In XML DOM, which method is used to create a new element node?

- A) createNode()
- B) createElement()
- C) newElement()
- D) createNewNode()

Answer: B) createElement()

Which property of the XML DOM is used to access the root element of an XML document?

- A) rootElement
- B) firstChild
- C) documentElement
- D) rootNode

Answer: C) documentElement

What is the purpose of the appendChild() method in XML DOM?

- A) To remove a child node
- B) To add a new child node
- C) To replace a child node
- D) To clone a child node

Answer: B) To add a new child node

Which method is used to retrieve a list of elements with a specified tag name in XML DOM?

- A) getElementsByTagName()
- B) getElementById()
- C) getElementsByTagName()
- D) getElementsByTagName()

Answer: A) getElementsByTagName()

In XML DOM, what does the nodeValue property return?

- A) The name of the node
- B) The value of the node
- C) The type of the node
- D) The parent of the node

Answer: B) The value of the node

Which of the following is a valid XML DOM node type?

- A) Element
- B) Attribute
- C) Text
- D) All of the above

Answer: D) All of the above

What does the removeChild() method do in XML DOM?

- A) Adds a new child node
- B) Removes a specified child node
- C) Replaces a child node
- D) Clones a child node

Answer: B) Removes a specified child node

Which method is used to replace a child node in XML DOM?

- A) replaceChild()
- B) swapChild()
- C) changeChild()
- D) updateChild()

Answer: A) replaceChild()

Absolutely! Here are some additional multiple-choice questions (MCQs) focused on XML parsing:

XML Parsing MCQs

Which of the following is a common XML parser for Java?

- A) SAX
- B) DOM
- C) JAXP
- D) All of the above

Answer: D) All of the above

What does SAX stand for in XML parsing?

- A) Simple API for XML
- B) Standard API for XML
- C) Simple Access for XML
- D) Standard Access for XML

Answer: A) Simple API for XML

Which XML parser reads the entire XML document and constructs a tree representation in memory?

- A) SAX
- B) DOM
- C) StAX
- D) Pull

Answer: B) DOM

Which of the following is a streaming XML parser?

- A) DOM
- B) SAX
- C) JDOM
- D) XOM

Answer: B) SAX

In XML parsing, what is the main advantage of using SAX over DOM?

- A) SAX is faster and uses less memory
- B) SAX is easier to use
- C) SAX supports XPath
- D) SAX can modify the XML document

Answer: A) SAX is faster and uses less memory

Which method is used to parse an XML document in a DOM parser?

- A) parseDocument()
- B) parseXML()
- C) parse()
- D) loadXML()

Answer: C) parse()

What is the main disadvantage of using DOM for XML parsing?

- A) It cannot handle large XML documents
- B) It is slower and uses more memory
- C) It does not support namespaces
- D) It cannot validate XML documents

Answer: B) It is slower and uses more memory

Which of the following is true about StAX parser?

- A) It is a pull parser
- B) It is a push parser
- C) It builds a tree structure in memory
- D) It does not support event-based parsing

Answer: A) It is a pull parser

Which method in SAX parser is called when the parser encounters the start of an element?

- A) startElement()
- B) beginElement()
- C) elementStart()
- D) onElementStart()

Answer: A) startElement()

In XML parsing, what does the term “well-formed” mean?

- A) The XML document adheres to the XML syntax rules
- B) The XML document is validated against a DTD or schema
- C) The XML document contains no errors
- D) The XML document is properly formatted for display

Answer: A) The XML document adheres to the XML syntax rules

JAXP MCQs

1. What does JAXP stand for?

- A) Java API for XML Parsing
- B) Java API for XML Processing
- C) Java Application for XML Parsing
- D) Java Application for XML Processing
- **Answer:** B) Java API for XML Processing

2. Which of the following is a component of JAXP?

- A) DOM
- B) SAX
- C) XSLT
- D) All of the above
- **Answer:** D) All of the above

3. What is the primary purpose of JAXP?

- A) To parse XML documents
- B) To transform XML documents
- C) To validate XML documents
- D) All of the above
- **Answer:** D) All of the above

4. Which JAXP interface is used to create a new instance of a DOM parser?

- A) DocumentBuilderFactory
- B) SAXParserFactory
- C) TransformerFactory
- D) XMLInputFactory
- **Answer:** A) DocumentBuilderFactory

5. Which method in JAXP is used to parse an XML document using a DOM parser?

- A) parseDocument()
- B) parseXML()
- C) parse()
- D) loadXML()
- **Answer:** C) parse()

6. Which JAXP class is used to perform XSLT transformations?

- A) Transformer
- B) TransformerFactory
- C) XSLTProcessor
- D) XSLTTransformer
- **Answer:** A) Transformer

7. In JAXP, which method is used to create a new instance of a SAX parser?

- A) newSAXParser()
- B) createSAXParser()
- C) newInstance()
- D) newParser()
- **Answer:** C) newInstance()

8. Which of the following is NOT a feature of JAXP?

- A) Platform independence

- B) Support for multiple XML parsers
 - C) Integration with Java Collections Framework
 - D) Support for XSLT transformations
 - **Answer:** C) Integration with Java Collections Framework
- 9. **What is the role of the DocumentBuilder class in JAXP?**
 - A) To build XML documents from scratch
 - B) To parse XML documents into a DOM tree
 - C) To transform XML documents using XSLT
 - D) To validate XML documents against a schema
 - **Answer:** B) To parse XML documents into a DOM tree
- 10. **Which JAXP interface is used to create a new instance of a Transformer?**
 - A) TransformerFactory
 - B) TransformerBuilder
 - C) TransformerCreator
 - D) TransformerInstance
 - **Answer:** A) TransformerFactory
- 11. **In JAXP, which method is used to set the output properties for a Transformer?**
 - A) setOutputProperties()
 - B) setOutput()
 - C) setProperties()
 - D) setOutputFormat()
 - **Answer:** A) setOutputProperties()
- 12. **Which JAXP class is used to validate an XML document against an XML Schema?**
 - A) Validator
 - B) SchemaValidator
 - C) XMLValidator
 - D) SchemaFactory
 - **Answer:** A) Validator
- 13. **What is the default parser used by JAXP if no specific parser is specified?**
 - A) Xerces
 - B) Crimson
 - C) Aelfred
 - D) Piccolo
 - **Answer:** A) Xerces
- 14. **Which method in JAXP is used to transform an XML document using a Transformer?**
 - A) transform()
 - B) applyTransformation()
 - C) executeTransformation()
 - D) performTransformation()
 - **Answer:** A) transform()
- 15. **In JAXP, which method is used to obtain a new instance of a SchemaFactory?**
 - A) newSchemaFactory()
 - B) createSchemaFactory()
 - C) newInstance()
 - D) getSchemaFactory()
 - **Answer:** C) newInstance()

XML Validation MCQs

1. **What is the purpose of XML validation?**
 - A) To check the syntax of the XML document
 - B) To ensure the XML document adheres to a defined structure
 - C) To transform the XML document
 - D) To parse the XML document
 - **Answer:** B) To ensure the XML document adheres to a defined structure
2. **Which of the following is used to define the structure and rules for an XML document?**
 - A) XML Schema
 - B) DTD (Document Type Definition)
 - C) XSLT

- D) Both A and B
 - **Answer:** D) Both A and B
- 3. **What does DTD stand for?**
 - A) Document Type Definition
 - B) Data Type Definition
 - C) Document Type Declaration
 - D) Data Type Declaration
 - **Answer:** A) Document Type Definition
- 4. **Which of the following is a benefit of using XML Schema over DTD?**
 - A) XML Schema supports data types
 - B) XML Schema is written in XML
 - C) XML Schema is more powerful and flexible
 - D) All of the above
 - **Answer:** D) All of the above
- 5. **Which JAXP class is used to validate an XML document against an XML Schema?**
 - A) Validator
 - B) SchemaValidator
 - C) XMLValidator
 - D) SchemaFactory
 - **Answer:** A) Validator
- 6. **In XML Schema, which element is used to define a simple type?**
 - A) <simpleType>
 - B) <element>
 - C) <complexType>
 - D) <type>
 - **Answer:** A) <simpleType>
- 7. **Which method in JAXP is used to set the schema for validation?**
 - A) setSchema()
 - B) setValidationSchema()
 - C) setSchemaFactory()
 - D) setValidatorSchema()
 - **Answer:** A) setSchema()
- 8. **What is the file extension for an XML Schema file?**
 - A) .xml
 - B) .xsd
 - C) .dtd
 - D) .xsl
 - **Answer:** B) .xsd
- 9. **Which of the following is NOT a valid XML Schema data type?**
 - A) xs:string
 - B) xs:integer
 - C) xs:date
 - D) xs:char
 - **Answer:** D) xs:char
- 10. **In XML validation, what does the term “valid” mean?**
 - A) The XML document adheres to the XML syntax rules
 - B) The XML document is well-formed and adheres to the rules defined by a DTD or Schema
 - C) The XML document contains no errors
 - D) The XML document is properly formatted for display
 - **Answer:** B) The XML document is well-formed and adheres to the rules defined by a DTD or Schema
- 11. **Which method in JAXP is used to validate an XML document against a schema?**
 - A) validate()
 - B) checkValidity()
 - C) parseAndValidate()
 - D) verify()

- **Answer:** A) validate()
- 12. **What is the purpose of the SchemaFactory class in JAXP?**
 - A) To create a new schema instance
 - B) To parse XML documents
 - C) To transform XML documents
 - D) To build a DOM tree
 - **Answer:** A) To create a new schema instance
- 13. **Which of the following is a valid XML Schema element for defining an attribute?**
 - A) <attribute>
 - B) [xs:attribute](#)
 - C) [xs:attr](#)
 - D) <attr>
 - **Answer:** B) [xs:attribute](#)
- 14. **In XML Schema, which element is used to define a complex type?**
 - A) <complexType>
 - B) <element>
 - C) <simpleType>
 - D) <type>
 - **Answer:** A) <complexType>
- 15. **Which of the following is true about DTD?**
 - A) DTD is written in XML
 - B) DTD supports namespaces
 - C) DTD is less powerful than XML Schema
 - D) DTD supports data types
 - **Answer:** C) DTD is less powerful than XML Schema

Servlets MCQs

1. **What is a servlet in Java?**
 - A) A server-side program that handles client requests and generates dynamic content
 - B) A client-side program that handles server requests
 - C) A database management system
 - D) A type of Java applet
 - **Answer:** A) A server-side program that handles client requests and generates dynamic content
2. **Which package contains the servlet classes and interfaces?**
 - A) javax.servlet
 - B) java.servlet
 - C) org.servlet
 - D) com.servlet
 - **Answer:** A) javax.servlet
3. **Which method is called when a servlet is first loaded into memory?**
 - A) init()
 - B) service()
 - C) doGet()
 - D) doPost()
 - **Answer:** A) init()
4. **Which method is used to handle HTTP GET requests in a servlet?**
 - A) doGet()
 - B) doPost()
 - C) service()
 - D) init()
 - **Answer:** A) doGet()
5. **Which method is used to handle HTTP POST requests in a servlet?**
 - A) doGet()
 - B) doPost()
 - C) service()
 - D) init()

- **Answer:** B) doPost()
- 6. **What is the lifecycle method that is called when a servlet is about to be destroyed?**
 - A) destroy()
 - B) terminate()
 - C) stop()
 - D) end()
 - **Answer:** A) destroy()
- 7. **Which interface must be implemented by all servlets?**
 - A) Servlet
 - B) HttpServlet
 - C) GenericServlet
 - D) ServletContext
 - **Answer:** A) Servlet
- 8. **Which method is used to retrieve initialization parameters in a servlet?**
 - A) getInitParameter()
 - B) getParameter()
 - C) getConfigParameter()
 - D) getServletParameter()
 - **Answer:** A) getInitParameter()
- 9. **What is the purpose of the service() method in a servlet?**
 - A) To initialize the servlet
 - B) To handle client requests
 - C) To destroy the servlet
 - D) To configure the servlet
 - **Answer:** B) To handle client requests
- 10. **Which object is used to send a response back to the client in a servlet?**
 - A) HttpServletResponse
 - B) HttpServletRequest
 - C) ServletContext
 - D) ServletConfig
 - **Answer:** A) HttpServletResponse
- 11. **Which method is used to forward a request from one servlet to another?**
 - A) forward()
 - B) sendRedirect()
 - C) include()
 - D) dispatch()
 - **Answer:** A) forward()
- 12. **Which method is used to redirect a request to a different URL?**
 - A) forward()
 - B) sendRedirect()
 - C) include()
 - D) dispatch()
 - **Answer:** B) sendRedirect()
- 13. **What is the purpose of the ServletContext interface?**
 - A) To provide information about the servlet's environment
 - B) To handle client requests
 - C) To manage servlet lifecycle
 - D) To configure the servlet
 - **Answer:** A) To provide information about the servlet's environment
- 14. **Which method is used to include the content of another resource in a servlet response?**
 - A) include()
 - B) forward()
 - C) sendRedirect()
 - D) dispatch()
 - **Answer:** A) include()
- 15. **Which of the following is true about the HttpServlet class?**

- A) It is an abstract class
 - B) It implements the Servlet interface
 - C) It provides methods to handle HTTP-specific services
 - D) All of the above
 - **Answer:** D) All of the above
16. Which method is used to set the content type of the response in a servlet?
- A) setContentTypes()
 - B) setType()
 - C) setResponseType()
 - D) setMimeType()
 - **Answer:** A) setContentTypes()
17. What is the default HTTP method handled by the service() method in a servlet?
- A) GET
 - B) POST
 - C) Both GET and POST
 - D) Neither GET nor POST
 - **Answer:** C) Both GET and POST
18. Which of the following is NOT a valid HTTP method that can be handled by a servlet?
- A) GET
 - B) POST
 - C) PUT
 - D) FETCH
 - **Answer:** D) FETCH
19. Which method is used to retrieve a parameter value from the request in a servlet?
- A) getParameter()
 - B) getRequestParameter()
 - C) getParam()
 - D) getRequestParam()
 - **Answer:** A) getParameter()
20. Which of the following is used to manage sessions in servlets?
- A) HttpSession
 - B) ServletSession
 - C) SessionManager
 - D) SessionContext
 - **Answer:** A) HttpSession

Sure! Here are some multiple-choice questions (MCQs) on JSP (JavaServer Pages) along with their answers:

JSP MCQs

1. What does JSP stand for?
 - A) JavaServer Pages
 - B) JavaScript Pages
 - C) Java Servlet Pages
 - D) JavaServer Programming
 - **Answer:** A) JavaServer Pages
2. Which tag is used to include a file in a JSP page?
 - A) [jsp:include](#)
 - B) [jsp:forward](#)
 - C) [jsp:import](#)
 - D) [jsp:insert](#)
 - **Answer:** A) [jsp:include](#)
3. Which of the following is a scripting element in JSP?
 - A) Declaration
 - B) Scriptlet
 - C) Expression
 - D) All of the above
 - **Answer:** D) All of the above
4. Which tag is used to forward a request to another resource in JSP?

- A) [jsp:forward](#)
 - B) [jsp:include](#)
 - C) [jsp:redirect](#)
 - D) [jsp:dispatch](#)
 - **Answer:** A) [jsp:forward](#)
- 5. **What is the file extension for a JSP file?**
 - A) .jsp
 - B) .java
 - C) .html
 - D) .xml
 - **Answer:** A) .jsp
- 6. **Which of the following is NOT a valid JSP directive?**
 - A) page
 - B) include
 - C) taglib
 - D) script
 - **Answer:** D) script
- 7. **Which method is used to handle HTTP GET requests in a JSP page?**
 - A) `_jspService()`
 - B) `doGet()`
 - C) `service()`
 - D) `init()`
 - **Answer:** A) `_jspService()`
- 8. **Which of the following is used to declare variables and methods in a JSP page?**
 - A) Declaration
 - B) Scriptlet
 - C) Expression
 - D) Directive
 - **Answer:** A) Declaration
- 9. **Which JSP tag is used to set the value of a JavaBean property?**
 - A) [jsp:setProperty](#)
 - B) [jsp:getProperty](#)
 - C) [jsp:useBean](#)
 - D) [jsp:property](#)
 - **Answer:** A) [jsp:setProperty](#)
- 10. **Which of the following is true about JSP expressions?**
 - A) They are evaluated and inserted into the output
 - B) They are used to declare variables
 - C) They are used to include other resources
 - D) They are used to forward requests
 - **Answer:** A) They are evaluated and inserted into the output
- 11. **Which of the following is a valid scope for a JavaBean in JSP?**
 - A) page
 - B) request
 - C) session
 - D) All of the above
 - **Answer:** D) All of the above
- 12. **Which JSP directive is used to include a file during the translation phase?**
 - A) include
 - B) page
 - C) taglib
 - D) script
 - **Answer:** A) include
- 13. **Which of the following is NOT a valid JSP action?**
 - A) [jsp:useBean](#)
 - B) [jsp:setProperty](#)
 - C) [jsp:getProperty](#)

- D) [jsp:script](#)
 - **Answer:** D) [jsp:script](#)
- 14. **What is the purpose of the page directive in JSP?**
 - A) To define page-dependent attributes
 - B) To include other resources
 - C) To declare variables and methods
 - D) To forward requests
 - **Answer:** A) To define page-dependent attributes
- 15. **Which of the following is used to handle exceptions in JSP?**
 - A) `errorPage`
 - B) `exceptionPage`
 - C) `errorHandler`
 - D) `exceptionHandler`
 - **Answer:** A) `errorPage`
- 16. **Which method is called when a JSP page is first loaded into memory?**
 - A) `jspInit()`
 - B) `_jspService()`
 - C) `jspDestroy()`
 - D) `init()`
 - **Answer:** A) `jspInit()`
- 17. **Which of the following is true about JSP scriptlets?**
 - A) They can contain any valid Java code
 - B) They are used to declare variables
 - C) They are used to include other resources
 - D) They are used to forward requests
 - **Answer:** A) They can contain any valid Java code
- 18. **Which JSP tag is used to retrieve the value of a JavaBean property?**
 - A) [jsp:getProperty](#)
 - B) [jsp:setProperty](#)
 - C) [jsp:useBean](#)
 - D) [jsp:property](#)
 - **Answer:** A) [jsp:getProperty](#)
- 19. **Which of the following is NOT a valid attribute of the page directive?**
 - A) `language`
 - B) `contentType`
 - C) `import`
 - D) `include`
 - **Answer:** D) `include`
- 20. **Which of the following is used to include dynamic content in a JSP page?**
 - A) `jsp:include`
 - B) `jsp:forward`
 - C) `jsp:import`
 - D) `jsp:insert`
 - **Answer:** A) `jsp:include`

JDBC MCQs

1. **What does JDBC stand for?**
 - A) Java Database Connectivity
 - B) Java Data Connection
 - C) Java Data Communication
 - D) Java Database Communication
 - **Answer:** A) Java Database Connectivity
2. **Which package contains the JDBC classes and interfaces?**
 - A) `java.jdbc`
 - B) `javax.jdbc`
 - C) `java.sql`
 - D) `javax.sql`
 - **Answer:** C) `java.sql`

3. **Which interface is used to establish a connection to a database in JDBC?**
- A) Connection
 - B) DriverManager
 - C) Statement
 - D) ResultSet
 - **Answer: A) Connection**
4. **Which method is used to execute a SQL query in JDBC?**
- A) executeQuery()
 - B) execute()
 - C) runQuery()
 - D) executeSQL()
 - **Answer: A) executeQuery()**
5. **Which interface provides methods to execute SQL statements in JDBC?**
- A) Statement
 - B) Connection
 - C) ResultSet
 - D) DriverManager
 - **Answer: A) Statement**
6. **Which method is used to execute an SQL update statement in JDBC?**
- A) executeUpdate()
 - B) execute()
 - C) runUpdate()
 - D) updateSQL()
 - **Answer: A) executeUpdate()**
7. **Which interface is used to retrieve the results of a SQL query in JDBC?**
- A) ResultSet
 - B) Statement
 - C) Connection
 - D) DriverManager
 - **Answer: A) ResultSet**
8. **Which method is used to close a database connection in JDBC?**
- A) close()
 - B) disconnect()
 - C) terminate()
 - D) end()
 - **Answer: A) close()**
9. **Which of the following is NOT a type of JDBC driver?**
- A) Type-1
 - B) Type-2
 - C) Type-3
 - D) Type-5
 - **Answer: D) Type-5**
10. **Which JDBC driver type is also known as the “thin driver”?**
- A) Type-1
 - B) Type-2
 - C) Type-3
 - D) Type-4
 - **Answer: D) Type-4**
11. **Which method is used to set the auto-commit mode in JDBC?**
- A) setAutoCommit()
 - B) setCommitMode()
 - C) autoCommit()
 - D) commitMode()
 - **Answer: A) setAutoCommit()**
12. **Which method is used to commit a transaction in JDBC?**
- A) commit()
 - B) executeCommit()

- C) runCommit()
 - D) finalizeCommit()
 - **Answer: A) commit()**
- 13. **Which method is used to roll back a transaction in JDBC?**
 - A) rollback()
 - B) executeRollback()
 - C) runRollback()
 - D) finalizeRollback()
 - **Answer: A) rollback()**
- 14. **Which interface is used to execute parameterized SQL queries in JDBC?**
 - A) PreparedStatement
 - B) Statement
 - C) CallableStatement
 - D) ResultSet
 - **Answer: A) PreparedStatement**
- 15. **Which method is used to execute a stored procedure in JDBC?**
 - A) execute()
 - B) executeProcedure()
 - C) runProcedure()
 - D) callProcedure()
 - **Answer: A) execute()**
- 16. **Which interface is used to call stored procedures in JDBC?**
 - A) CallableStatement
 - B) PreparedStatement
 - C) Statement
 - D) ResultSet
 - **Answer: A) CallableStatement**
- 17. **Which method is used to retrieve metadata about the database in JDBC?**
 - A) getMetaData()
 - B) getDatabaseMetaData()
 - C) getDBMetaData()
 - D) getMetaInfo()
 - **Answer: B) getDatabaseMetaData()**
- 18. **Which method is used to retrieve metadata about the result set in JDBC?**
 - A) getMetaData()
 - B) getResultSetMetaData()
 - C) getRSMetaData()
 - D) getMetaInfo()
 - **Answer: A) getMetaData()**
- 19. **Which method is used to retrieve the next row from a ResultSet in JDBC?**
 - A) next()
 - B) getNext()
 - C) fetchNext()
 - D) moveNext()
 - **Answer: A) next()**
- 20. **Which method is used to retrieve a column value from a ResultSet in JDBC?**
 - A) getString()
 - B) getInt()
 - C) getObject()
 - D) All of the above
 - **Answer: D) All of the above**
- 21. **Which method is used to check if a ResultSet is empty in JDBC?**
 - A) isEmpty()
 - B) isBeforeFirst()
 - C) isAfterLast()
 - D) isNull()
 - **Answer: B) isBeforeFirst()**

22. Which method is used to move the cursor to the first row in a ResultSet in JDBC?
- A) first()
 - B) moveToFirst()
 - C) goToFirst()
 - D) setFirst()
 - **Answer: A) first()**
23. Which method is used to move the cursor to the last row in a ResultSet in JDBC?
- A) last()
 - B) moveToLast()
 - C) goToLast()
 - D) setLast()
 - **Answer: A) last()**
24. Which method is used to move the cursor to a specific row in a ResultSet in JDBC?
- A) absolute()
 - B) moveToRow()
 - C) goToRow()
 - D) setRow()
 - **Answer: A) absolute()**
25. Which method is used to move the cursor to the previous row in a ResultSet in JDBC?
- A) previous()
 - B) moveToPrevious()
 - C) goToPrevious()
 - D) setPrevious()
 - **Answer: A) previous()**

Web Services MCQs

1. What is a web service?
- A) A software system designed to support interoperable machine-to-machine interaction over a network
 - B) A web page that provides information to users
 - C) A service provided by a web server to host websites
 - D) A tool for designing web pages
 - **Answer: A) A software system designed to support interoperable machine-to-machine interaction over a network**
2. Which protocol is commonly used for web services?
- A) HTTP
 - B) FTP
 - C) SMTP
 - D) POP3
 - **Answer: A) HTTP**
3. What does SOAP stand for?
- A) Simple Object Access Protocol
 - B) Simple Object Application Protocol
 - C) Service Oriented Access Protocol
 - D) Service Object Application Protocol
 - **Answer: A) Simple Object Access Protocol**
4. Which language is used to describe the services offered by a web service?
- A) WSDL
 - B) XML
 - C) HTML
 - D) JSON
 - **Answer: A) WSDL**
5. What does WSDL stand for?
- A) Web Services Description Language
 - B) Web Services Definition Language
 - C) Web Services Deployment Language
 - D) Web Services Data Language
 - **Answer: A) Web Services Description Language**

6. Which of the following is a RESTful web service constraint?
- A) Statelessness
 - B) Stateful interactions
 - C) Synchronous communication
 - D) Tight coupling
 - **Answer: A) Statelessness**
7. What does REST stand for?
- A) Representational State Transfer
 - B) Remote State Transfer
 - C) Representational Service Transfer
 - D) Remote Service Transfer
 - **Answer: A) Representational State Transfer**
8. Which format is commonly used for data exchange in RESTful web services?
- A) JSON
 - B) XML
 - C) HTML
 - D) Both A and B
 - **Answer: D) Both A and B**
9. Which HTTP method is used to retrieve data from a web service?
- A) GET
 - B) POST
 - C) PUT
 - D) DELETE
 - **Answer: A) GET**
10. Which HTTP method is used to create a new resource in a web service?
- A) POST
 - B) GET
 - C) PUT
 - D) DELETE
 - **Answer: A) POST**
11. Which HTTP method is used to update an existing resource in a web service?
- A) PUT
 - B) GET
 - C) POST
 - D) DELETE
 - **Answer: A) PUT**
12. Which HTTP method is used to delete a resource in a web service?
- A) DELETE
 - B) GET
 - C) POST
 - D) PUT
 - **Answer: A) DELETE**
13. What is the purpose of UDDI in web services?
- A) To provide a directory for storing information about web services
 - B) To describe the services offered by a web service
 - C) To exchange data between web services
 - D) To secure web services
 - **Answer: A) To provide a directory for storing information about web services**
14. What does UDDI stand for?
- A) Universal Description, Discovery, and Integration
 - B) Universal Data Description and Integration
 - C) Unified Data Description and Integration
 - D) Unified Description, Discovery, and Integration
 - **Answer: A) Universal Description, Discovery, and Integration**
15. Which of the following is a key characteristic of RESTful web services?
- A) Statelessness
 - B) Tight coupling

- ☐ C) Synchronous communication
 - ☐ D) Platform dependence
 - ☐ **Answer: A) Statelessness**
16. Which of the following is NOT a valid HTTP status code for a successful request?
- ☐ A) 200 OK
 - ☐ B) 201 Created
 - ☐ C) 204 No Content
 - ☐ D) 404 Not Found
 - ☐ **Answer: D) 404 Not Found**
17. Which of the following is a security protocol commonly used with web services?
- ☐ A) SSL/TLS
 - ☐ B) FTP
 - ☐ C) SMTP
 - ☐ D) POP3
 - ☐ **Answer: A) SSL/TLS**
18. Which of the following is a common tool for testing web services?
- ☐ A) Postman
 - ☐ B) Eclipse
 - ☐ C) NetBeans
 - ☐ D) IntelliJ IDEA
 - ☐ **Answer: A) Postman**
19. Which of the following is a benefit of using web services?
- ☐ A) Interoperability
 - ☐ B) Scalability
 - ☐ C) Reusability
 - ☐ D) All of the above
 - ☐ **Answer: D) All of the above**
20. Which of the following is NOT a type of web service?
- ☐ A) SOAP
 - ☐ B) REST
 - ☐ C) FTP
 - ☐ D) XML-RPC
 - ☐ **Answer: C) FTP**
21. Which of the following is a lightweight protocol for exchanging information in a decentralized, distributed environment?
- ☐ A) JSON-RPC
 - ☐ B) SOAP
 - ☐ C) REST
 - ☐ D) XML-RPC
 - ☐ **Answer: C) REST**
22. Which of the following is a markup language used to encode documents in a format that is both human-readable and machine-readable?
- ☐ A) XML
 - ☐ B) HTML
 - ☐ C) JSON
 - ☐ D) YAML
 - ☐ **Answer: A) XML**
23. Which of the following is a common format for representing structured data based on JavaScript object syntax?
- ☐ A) JSON
 - ☐ B) XML
 - ☐ C) HTML
 - ☐ D) YAML
 - ☐ **Answer: A) JSON**
24. Which of the following is a protocol used for accessing and managing web services?
- ☐ A) HTTP
 - ☐ B) FTP

- ☐ C) SMTP
 - ☐ D) POP3
 - ☐ **Answer: A) HTTP**
25. Which of the following is a standard for describing the capabilities of a web service?
- ☐ A) WSDL
 - ☐ B) XML
 - ☐ C) JSON
 - ☐ D) HTML
 - ☐ **Answer: A) WSDL**

More Web Services MCQs

1. Which of the following is a key principle of RESTful web services?
 - ☐ A) Statelessness
 - ☐ B) Stateful interactions
 - ☐ C) Tight coupling
 - ☐ D) Synchronous communication
 - ☐ **Answer: A) Statelessness**
2. Which of the following is a common HTTP status code for a successful POST request?
 - ☐ A) 200 OK
 - ☐ B) 201 Created
 - ☐ C) 204 No Content
 - ☐ D) 404 Not Found
 - ☐ **Answer: B) 201 Created**
3. Which of the following is NOT a valid HTTP status code for a client error?
 - ☐ A) 400 Bad Request
 - ☐ B) 401 Unauthorized
 - ☐ C) 403 Forbidden
 - ☐ D) 500 Internal Server Error
 - ☐ **Answer: D) 500 Internal Server Error**
4. Which of the following is a lightweight data-interchange format that is easy for humans to read and write?
 - ☐ A) JSON
 - ☐ B) XML
 - ☐ C) HTML
 - ☐ D) YAML
 - ☐ **Answer: A) JSON**
5. Which of the following is a common tool for creating and testing RESTful web services?
 - ☐ A) Postman
 - ☐ B) Eclipse
 - ☐ C) NetBeans
 - ☐ D) IntelliJ IDEA
 - ☐ **Answer: A) Postman**
6. Which of the following is a benefit of using RESTful web services?
 - ☐ A) Scalability
 - ☐ B) Interoperability
 - ☐ C) Flexibility
 - ☐ D) All of the above
 - ☐ **Answer: D) All of the above**
7. Which of the following is a common security mechanism for securing web services?
 - ☐ A) SSL/TLS
 - ☐ B) FTP
 - ☐ C) SMTP
 - ☐ D) POP3
 - ☐ **Answer: A) SSL/TLS**
8. Which of the following is a common format for representing structured data based on JavaScript object syntax?
 - ☐ A) JSON
 - ☐ B) XML

- C) HTML
 - D) YAML
 - **Answer: A) JSON**
- 9. **Which of the following is a protocol used for accessing and managing web services?**
 - A) HTTP
 - B) FTP
 - C) SMTP
 - D) POP3
 - **Answer: A) HTTP**
- 10. **Which of the following is a standard for describing the capabilities of a web service?**
 - A) WSDL
 - B) XML
 - C) JSON
 - D) HTML
 - **Answer: A) WSDL**
- 11. **Which of the following is a key characteristic of SOAP-based web services?**
 - A) Platform independence
 - B) Tight coupling
 - C) Synchronous communication
 - D) Statelessness
 - **Answer: A) Platform independence**
- 12. **Which of the following is a common use case for web services?**
 - A) Integrating different applications
 - B) Exchanging data between systems
 - C) Providing APIs for third-party developers
 - D) All of the above
 - **Answer: D) All of the above**
- 13. **Which of the following is a common HTTP status code for a successful GET request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found
 - **Answer: A) 200 OK**
- 14. **Which of the following is a common HTTP status code for a successful DELETE request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found
 - **Answer: C) 204 No Content**
- 15. **Which of the following is a common HTTP status code for a server error?**
 - A) 500 Internal Server Error
 - B) 400 Bad Request
 - C) 401 Unauthorized
 - D) 403 Forbidden
 - **Answer: A) 500 Internal Server Error**
- 16. **Which of the following is a common HTTP status code for a client error?**
 - A) 400 Bad Request
 - B) 500 Internal Server Error
 - C) 200 OK
 - D) 201 Created
 - **Answer: A) 400 Bad Request**
- 17. **Which of the following is a common HTTP status code for a successful PUT request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found

- **Answer:** A) 200 OK
- 18. **Which of the following is a common HTTP status code for a successful PATCH request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found
 - **Answer:** A) 200 OK
- 19. **Which of the following is a common HTTP status code for a successful HEAD request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found
 - **Answer:** A) 200 OK
- 20. **Which of the following is a common HTTP status code for a successful OPTIONS request?**
 - A) 200 OK
 - B) 201 Created
 - C) 204 No Content
 - D) 404 Not Found
 - **Answer:** A) 200 OK

Web Services Architecture MCQs

1. **What is the primary purpose of web services architecture?**
 - A) To enable communication between different software applications over a network
 - B) To design web pages
 - C) To manage databases
 - D) To create desktop applications
 - **Answer:** A) To enable communication between different software applications over a network
2. **Which of the following is a core component of web services architecture?**
 - A) Service provider
 - B) Service consumer
 - C) Service registry
 - D) All of the above
 - **Answer:** D) All of the above
3. **What is the role of the service provider in web services architecture?**
 - A) To publish and provide web services
 - B) To consume web services
 - C) To store information about web services
 - D) To secure web services
 - **Answer:** A) To publish and provide web services
4. **What is the role of the service consumer in web services architecture?**
 - A) To consume web services
 - B) To publish web services
 - C) To store information about web services
 - D) To secure web services
 - **Answer:** A) To consume web services
5. **What is the role of the service registry in web services architecture?**
 - A) To store information about web services
 - B) To publish web services
 - C) To consume web services
 - D) To secure web services
 - **Answer:** A) To store information about web services
6. **Which of the following is a standard protocol used for web services communication?**
 - A) SOAP
 - B) HTTP
 - C) FTP

- D) Both A and B
 - **Answer:** D) Both A and B
- 7. **Which of the following is a key characteristic of SOAP-based web services?**
 - A) Platform independence
 - B) Tight coupling
 - C) Synchronous communication
 - D) Statelessness
 - **Answer:** A) Platform independence
- 8. **Which of the following is a key characteristic of RESTful web services?**
 - A) Statelessness
 - B) Tight coupling
 - C) Synchronous communication
 - D) Platform dependence
 - **Answer:** A) Statelessness
- 9. **Which language is used to describe the services offered by a web service?**
 - A) WSDL
 - B) XML
 - C) HTML
 - D) JSON
 - **Answer:** A) WSDL
- 10. **What does WSDL stand for?**
 - A) Web Services Description Language
 - B) Web Services Definition Language
 - C) Web Services Deployment Language
 - D) Web Services Data Language
 - **Answer:** A) Web Services Description Language
- 11. **Which of the following is a benefit of using web services?**
 - A) Interoperability
 - B) Scalability
 - C) Reusability
 - D) All of the above
 - **Answer:** D) All of the above
- 12. **Which of the following is a security protocol commonly used with web services?**
 - A) SSL/TLS
 - B) FTP
 - C) SMTP
 - D) POP3
 - **Answer:** A) SSL/TLS
- 13. **Which of the following is a common tool for testing web services?**
 - A) Postman
 - B) Eclipse
 - C) NetBeans
 - D) IntelliJ IDEA
 - **Answer:** A) Postman
- 14. **Which of the following is a common format for representing structured data based on JavaScript object syntax?**
 - A) JSON
 - B) XML
 - C) HTML
 - D) YAML
 - **Answer:** A) JSON
- 15. **Which of the following is a protocol used for accessing and managing web services?**
 - A) HTTP
 - B) FTP
 - C) SMTP
 - D) POP3
 - **Answer:** A) HTTP

16. Which of the following is a standard for describing the capabilities of a web service?

- A) WSDL
- B) XML
- C) JSON
- D) HTML
- **Answer: A) WSDL**

17. Which of the following is a key principle of RESTful web services?

- A) Statelessness
- B) Stateful interactions
- C) Tight coupling
- D) Synchronous communication
- **Answer: A) Statelessness**

SUBJECTIVE

Module 12: XML

1. **What is XML and why is it used?** XML (eXtensible Markup Language) is used to store and transport data, allowing for a flexible way to create information formats and electronically share structured data.
2. **How does XML differ from HTML?** XML is designed to transport and store data, while HTML is designed to display data.
3. **What are XML namespaces?** XML namespaces provide a method to avoid element name conflicts by qualifying names with a unique identifier.
4. **What is an XML Schema?** An XML Schema defines the structure and data types of an XML document.
5. **How do you validate an XML document?** An XML document is validated against a DTD (Document Type Definition) or an XML Schema to ensure it conforms to the defined structure.

Module 13: XML DOM

1. **What is the XML DOM?** The XML DOM (Document Object Model) is a programming interface for XML documents, representing the document as a tree structure.
2. **How do you access elements in the XML DOM?** Elements in the XML DOM can be accessed using methods like getElementById, getElementsByTagName, and getElementsByClassName.
3. **What is the purpose of the createElement method in XML DOM?** The createElement method is used to create a new element node in the XML DOM.
4. **How do you modify an XML DOM element?** You can modify an XML DOM element using methods like setAttribute, appendChild, and removeChild.
5. **What is the role of the loadXML method?** The loadXML method is used to load an XML document into the DOM.

Module 14: Introduction to JAXP

1. **What does JAXP stand for?** JAXP stands for Java API for XML Processing.
2. **What are the main components of JAXP?** The main components of JAXP are the SAX (Simple API for XML), DOM (Document Object Model), and XSLT (Extensible Stylesheet Language Transformations) APIs.
3. **How does JAXP support XML parsing?** JAXP supports XML parsing through SAX and DOM parsers.
4. **What is the role of the Transformer class in JAXP?** The Transformer class is used to transform XML documents using XSLT.
5. **How do you configure a JAXP parser?** A JAXP parser is configured using factory classes like DocumentBuilderFactory and SAXParserFactory.

Module 15: Servlets

1. **What is a servlet?** A servlet is a Java program that runs on a server and handles client requests, typically HTTP requests.
2. **How do you create a servlet?** A servlet is created by extending the HttpServlet class and overriding methods like doGet and doPost.
3. **What is the servlet lifecycle?** The servlet lifecycle consists of initialization (init), handling requests (service), and termination (destroy).

4. **How do you configure a servlet in a web application?** A servlet is configured in the web.xml file or using annotations like @WebServlet.
5. **What is the purpose of the doGet and doPost methods?** The doGet method handles HTTP GET requests, while the doPost method handles HTTP POST requests.

Module 16: Java Database Connectivity (JDBC)

1. **What is JDBC?** JDBC (Java Database Connectivity) is an API that allows Java applications to interact with databases.
2. **How do you establish a JDBC connection?** A JDBC connection is established using the DriverManager.getConnection method with a database URL, username, and password.
3. **What is the role of the Statement interface in JDBC?** The Statement interface is used to execute SQL queries and retrieve results.
4. **How do you handle SQL exceptions in JDBC?** SQL exceptions are handled using try-catch blocks and the SQLException class.
5. **What is a PreparedStatement and how is it used?** A PreparedStatement is a precompiled SQL statement that can be executed multiple times with different parameters.

Module 17: Introduction to Web Services

1. **What are web services?** Web services are standardized ways of integrating web-based applications using XML, SOAP, WSDL, and UDDI open standards over an internet protocol backbone.
2. **What is SOAP?** SOAP (Simple Object Access Protocol) is a protocol for exchanging structured information in web services.
3. **What is WSDL?** WSDL (Web Services Description Language) is an XML-based language for describing web services and how to access them.
4. **What is the role of UDDI in web services?** UDDI (Universal Description, Discovery, and Integration) is a directory service for discovering web services.
5. **How do web services communicate?** Web services communicate over the internet using protocols like HTTP and SOAP.

Module 18: Web Services Architecture

1. **What are the main components of web services architecture?** The main components are the service provider, service requester, and service registry.
2. **What is the role of the service provider in web services architecture?** The service provider hosts the web service and makes it available to clients.
3. **How does the service requester interact with web services?** The service requester finds and invokes web services using the service registry.
4. **What is the purpose of the service registry?** The service registry stores information about available web services and their interfaces.
5. **How do web services ensure interoperability?** Web services ensure interoperability through the use of open standards like XML, SOAP, and WSDL.

Module 19: Java Web Services

1. **What is JAX-WS?** JAX-WS (Java API for XML Web Services) is a Java API for creating and consuming web services.
2. **How do you create a JAX-WS web service?** A JAX-WS web service is created by annotating a Java class with @WebService and defining methods with @WebMethod.
3. **What is the role of the Endpoint class in JAX-WS?** The Endpoint class is used to publish a web service endpoint.
4. **How do you consume a JAX-WS web service?** A JAX-WS web service is consumed by generating a client proxy using tools like wsimport.
5. **What is the purpose of the @WebService annotation?** The @WebService annotation marks a Java class as a web service endpoint.